

From Replayable Traces to Provenance-Aware Rule Fusion: Auditable Reinforcement Learning at the Rule Level

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Abstract

Reinforcement learning (RL) policies are typically delivered as opaque neural checkpoints, and the logs of a training run verify that training happened rather than explaining what the policy learned. We study whether a population of independently trained policy networks can be described and composed at the level of discrete behavioral rules, and under which conditions that composition is trustworthy. We introduce a protocol in which (i) agents share one frozen, calibration-derived symbolizer; (ii) a passive observer converts each agent’s logged transitions into a bank of confidence-scored condition–action rules; (iii) an append-only hash-bound ledger and exact replay verifier certify, relative to the recorded artifact and the recorded environment version, that the recorded trace is the trace executed; and (iv) rules from multiple seeds are fused offline by a confidence-ranked arbitration policy with an explicit fallback for blind spots. On CartPole-v1, rule-level fusion reaches a mean held-out return of 205.96 over 100 common reset seeds versus 163.65 for a single-actor comparator selected post hoc by training return, a paired difference of +42.31 (95% bootstrap interval [23.98, 61.93]) that we treat as exploratory because the comparator was chosen after training and the task exhibits ceiling effects: on the same matched episodes four of the eight source actors reach the 500-step ceiling (mean and median over the eight actors 330.86 and 358.81), the actor-logit ensemble reaches 500.0, and a fitted-Q generalized-policy-improvement (GPI) baseline collapses to 9.34. The same protocol fails on Acrobot-v1: 89.2% of fusion decisions conflict across seeds, confidence-ranked arbitration returns −443.37, and random arbitration among candidate rules returns −187.02 on matched episodes—an open mechanism we dissect with a 200-state Monte-Carlo action-semantics probe (22.5% agreement with the best first action). Rule-set overlap across seeds (mean pairwise Jaccard 0.56) does not transfer to behavioral agreement on fresh states (51.4%, indistinguishable from chance), and a fitted-Q GPI baseline collapses in both environments (9.34 and −499.57). We report the protocol’s failure-driven design history—lossy grammar compression, rule-constrained training interference, confidence-threshold collapse, and serialization failures—as part of the scientific record, and bound the claims accordingly.

We resist the temptation to collapse *auditability* into a single score. The word bundles at least six separately testable predicates—trace integrity, lossless coding, rule coverage, behavioural agreement, composition quality, and value-model reliability—and *passing one test does not imply passing another*. This paper reports a bounded study that measures each predicate on its own terms and states every prediction before seeing any result.

1 Introduction

Reinforcement learning systems are increasingly deployed where their decisions matter, and a natural question accompanies every deployment: *what, exactly, has the trained agent learned to do?*

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Table 1: Auditability disaggregated into six separately measured predicates. The right column is a boundary each predicate enforces; none of them licenses the others.

Predicate	Estimand	What it does <i>not</i> establish
Trace integrity	hash chain + exact replay	causal explanation
Lossless coding	exact reconstruction of the trace	policy explanation; storage win
Rule coverage	held-out coverage of induced rules	correctness of the policy
Behavioural agreement	fresh-state action agreement	a shared policy core
Composition quality	return + decision mix	optimal arbitration
Value-model reliability	held-out value diagnostics	valid GPI composition

For a policy network, the honest answer usually requires replaying checkpoints and reading log statistics. Both are useful, and both stop short of explanation: a checkpoint is an opaque function from observations to actions, and a training log is evidence that a procedure ran, not a description of the behavior it produced.

1.1 Auditability is not a single property

The word *auditability* is convenient and dangerous: it suggests a single grade that a system either earns or does not. Our study instead treats it as six disjoint predicates, each with its own estimand and its own negative result. Collapsing them into one number would hide the fact that the same system can pass some and fail others.

We distinguish six *audit predicates* from three *verification mechanisms*. The predicates are trace integrity, lossless coding, rule coverage, behavioural agreement, composition quality, and value-model reliability. Hash-chain integrity, environment replay, and provenance resolution are verification mechanisms: they supply evidence for selected predicates, but they are not themselves predicates and are not interchangeable with one another. Keeping the two categories apart is what prevents a passing mechanism check from being reported as a passing predicate.

This decomposition is the spine of the paper. Section 2 states the predicates formally; Section 5 shows which design intuitions failed each one; and Section 6 reports a pass/fail verdict per predicate. The central negative result is that *rule overlap* (a coverage predicate) carries no information about *behavioural agreement* (a different predicate), so a high score on one must never be read as a high score on the other.

This gap matters most when several trained agents are available and we want to combine them. If a population of independently initialized policies each mastered the same task in a different way, can we compare their behaviors, identify what they agree on, and assemble a single policy that inherits the agreement? Three obstacles appear immediately. First, two policies with identical returns may behave differently on the same states, so return statistics cannot stand in for behavioral descriptions. Second, if we discretize states in a way that is private to each agent, the resulting descriptions are not comparable across agents. Third, using induced rules to *constrain* a policy during training is known—and, as we show, empirically—to distort the very behavior we hope to describe: an overly restrictive action mask trades exploration for compliance and can return a suboptimal policy [1], masking alters the policy gradient rather than merely restricting the action set [2], and injecting external knowledge into the exploration process changes what the learner is able to discover [3].

Our objective in this paper is therefore not another method for squeezing out higher returns. It is to turn trained policies into auditable, composable artifacts, and to measure precisely where that transformation succeeds and where it breaks. We pursue a rule-level description: a policy is

reported as a bank of condition–action rules over a *shared* discretized state space, together with an execution log whose fidelity is operationally verified and that ties every recorded decision to the exact environment transition that produced it.

The path from that intuition to a working protocol is a sequence of failures, and we treat those failures as evidence rather than obstacles. Early attempts at grammar-based compression of behavior turned out to be lossy; using induced rules to constrain training improved some seeds and damaged others; a high-confidence rule admission threshold produced policies with 100% blind spots; compression of the audit log increased its size; and a contaminated ledger schema had to be repaired by redefining the hashable core of a training record. Each failure changed the protocol, and each change is documented with its cause.

The main protocol that survived this process has four components. A frozen shared state symbolizer maps every agent’s observations into a common discrete vocabulary. A passive rule observer converts logged transitions into confidence-scored rules. A replay verifier and a hash-bound ledger certify that the logs are faithful. And an offline fusion policy arbitrates the rule banks of multiple seeds under an explicit blind-spot fallback. We evaluate the protocol on two control benchmarks with eight seeds each.

Our contributions are bounded and deliberately conservative. (1) Under the shared symbolizer, rule-level fusion of eight independently trained actors exceeded the selected single-actor comparator on the matched CartPole-v1 evaluation set by +42.31 return points (95% bootstrap interval [23.98, 61.93]) while remaining auditable at the artifact level (replay and provenance checks pass under the evaluated setup); the comparator was selected post hoc by training return and the task exhibits ceiling effects, so we report this comparison as exploratory rather than as a pre-declared win. We make the sensitivity of that headline explicit: on the same matched episodes four of the eight individual actors reach the ceiling (500.0) and fusion is 294.04 points below the best held-out actor, so the result is a statement about the selection rule rather than about fusion being a strong single policy. (2) On Acrobot-v1 the same mechanism fails in an instructive way: conflicts dominate, confidence-ranked arbitration underperforms random arbitration, and a Monte-Carlo action-semantics probe shows the fusion policy agrees with the best first action on only 22.5% of fresh states. (3) We contribute negative evidence that constrains the whole approach: rule-set overlap does not imply behavioral agreement (51.4% agreement on fresh states despite a pairwise Jaccard overlap of 0.56), and a fitted-Q GPI baseline collapses in both environments, which places strict limits on claiming that rule fusion is “better than value-based composition.”

The paper proceeds from the simple question to the full protocol. Section 2 fixes notation and states the audit problem. Section 4 describes the final protocol. Section 5 reconstructs the failure-driven design history that produced it. Section 6 reports the experiments, including all negative results. Sections 3–8 position the work, discuss it, and conclude.

2 Preliminaries and Problem Statement

2.1 Markov decision process and agents

We consider a finite-horizon Markov decision process (MDP) $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with state space $\mathcal{S} \subseteq \mathbb{R}^d$, discrete action space $\mathcal{A} = \{1, \dots, K\}$, transition kernel P , bounded reward r , and discount $\gamma \in (0, 1]$. A policy $\pi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$ maps states to action distributions. The discounted return of a trajectory is

$$G_t = \sum_{\tau=t}^T \gamma^{\tau-t} r(s_\tau, a_\tau), \quad (1)$$

and the value of a policy is its expected return from the initial state distribution.

We train agents with REINFORCE [4], the policy-gradient update

$$\nabla J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[(G_t - b(s_t)) \nabla_\theta \log \pi_\theta(a_t | s_t) \right], \quad (2)$$

where θ parameterizes the policy network and $b(s_t)$ is a learned state-value baseline. Each agent is a two-hidden-layer network with 64 units per layer; eight independent seeds are used throughout.

2.2 The audit problem

Given a trained deterministic policy π (the argmax of a network), an auditor can observe its behavior by executing it in the environment and recording transitions $(s, a, r, s', \text{term}, \text{trunc})$. We call the resulting artifact a *trace*. Three questions define our problem:

1. *Fidelity*. Can the auditor certify, without trusting the training harness, that the trace is exactly what was executed, and that a checkpoint reproduces the logged decisions?
2. *Description*. Can the trace be compressed into a finite, inspectable set of rules such that the rules are comparable across agents that were trained independently?
3. *Composition*. Can rule banks from several agents be fused into one policy, and under which conditions does the fused policy inherit the strengths—rather than the disagreements—of its sources?

Scope distinction. The three questions above are frequently conflated with stronger claims they do not support. Re-executing recorded transitions verifies the record. It does not recover policy weights over time, optimizer state, gradients, or the random-number state that produced them, and it cannot establish that the recorder was honest. We therefore keep four evidentiary layers strictly separate: *trace fidelity* (the record is what ran), *policy description* (the rule bank is a finite summary), *behavioural agreement* (two policies act alike on fresh states), *provenance* (which rule and source produced a decision), and *causal explanation* (why a decision was made). A result at one layer is never cited as evidence at another.

2.3 Scope and non-claims

To pre-empt over-reading, we state up front what this study does *not* claim. We do not claim general or complete auditability of arbitrary RL training; that symbolic overlap entails behavioural consensus; that induced rules are human-readable explanations of a neural policy; a storage-compression advantage over generic compression; optimal or mechanism-explained arbitration; validated value-model composition; or a natural-language semantics for the discrete communication symbols. Every positive claim below is bounded to two benchmark tasks, eight seeds per task, one shared quantizer per task, and one software/hardware configuration.

Throughout, we distinguish three evidentiary layers: *identification-bearing* measurements (behavioral agreement on fresh states), *diagnostic* measurements (rule statistics, fitted-Q diagnostics), and *operational-health* checks (hash-chain validity, replay equality). A perfect score on an operational-health check certifies the artifact; it does not certify the policy.

3 Related Work

We organize the positioning around three axes, following the most rigorous framing in the literature: (A) *assumption*—what each related line assumes about the policy or the environment; (B) *demonstrated achievement*—what it actually shows; (C) *conditions of failure*—where its guarantee breaks; and (D) *relationship to the present work*. This structure is adopted deliberately so that each related method is judged by the same four questions we ask of our own protocol.

Policy description and interpretability of RL. Post-hoc interpretability of policies typically produces saliency or attribution maps rather than inspectable behavioral rules, and such maps do not certify behavior on unseen states. Rule extraction for RL has a long line of work using decision trees and program synthesis. A closer line of work extracts structured policies from neural controllers: VIPER uses imitation-style policy extraction guided by a neural policy and its value estimates, with formal verification results on several control tasks [5], and other work directly optimizes interpretable decision-tree policies or controls the performance–complexity trade-off during policy learning [6]. Our setting differs in objective: we do not claim that the rule bank is a behaviorally equivalent replacement for the neural policy; instead we evaluate trace anchoring, cross-seed comparability, provenance, and the failure of rule overlap to predict behavior. The present study also differs in that rules are induced from a *shared, frozen symbolizer* so that rule banks from different seeds are directly comparable, and in that every rule is anchored to a replayed, hash-bound trace.

Value-based composition. Policy improvement theorems [7] show how a value function can certify improvement over a base policy, and generalized policy improvement (GPI) combines several value functions; formal GPI composes policies through approximate action-value functions and provides a performance bound that depends on value-estimation error [8]. Our rule-level arbiter is *not* an instance of theorem-backed GPI: it ranks candidate actions by empirical rule confidence and support rather than by estimates of source-policy action values Q^{π_i} , so no GPI-style policy-improvement guarantee applies to rule fusion. We therefore use “GPI baseline” only for the separate fitted-Q comparator (17), and we make no GPI-style policy-improvement claim for rule fusion. That comparator fails precisely because the fitted critics are nearly constant, a failure mode documented for FQE under off-policy coverage limits [9, 10] and consistent with the broader offline-RL problem of distribution shift and extrapolation error: a critic trained from a fixed replay distribution can assign unreliable values to state–action pairs induced by the composed policy [11, 12]. We read that failure as a validity diagnostic for this implementation, not as evidence against value-based composition in general. We do not position rule fusion as a replacement for value-based composition; the evidence supports only the bounded claim that, under a shared symbolizer, rule-level arbitration can exceed the selected single-actor comparator in a low-conflict regime.

Discrete symbolic communication and auditability. Emergent communication can appear under cooperative objectives [13, 14], but measurement is fraught: Lowe et al. [15] catalog pitfalls in which apparent communication is an artifact of the objective or the analysis. The “AI Mother Tongue” line shows VQ-VAE-derived discrete symbols can serve as an endogenous channel [16], and its audit agenda asks whether coordination logic can be reconstructed from discrete-symbol logs without hidden channels. The present work shares the audit agenda but does not rely on emergent communication: our symbols are a fixed calibration-derived abstraction, not a learned codebook, so the two lines are complementary rather than competing. We do not claim any result about learned symbol systems. Our causal-symbol test (Section 6.8) is a positive demonstration that the causal question *can* be answered by intervention in a designed task, complementing Lowe et al.’s catalog of ways correlation misleads.

Auditability of training records. Ledgering and replay verification of training runs are reproducibility practices. Our contribution on this axis is limited but explicit: a hash-bound ledger over

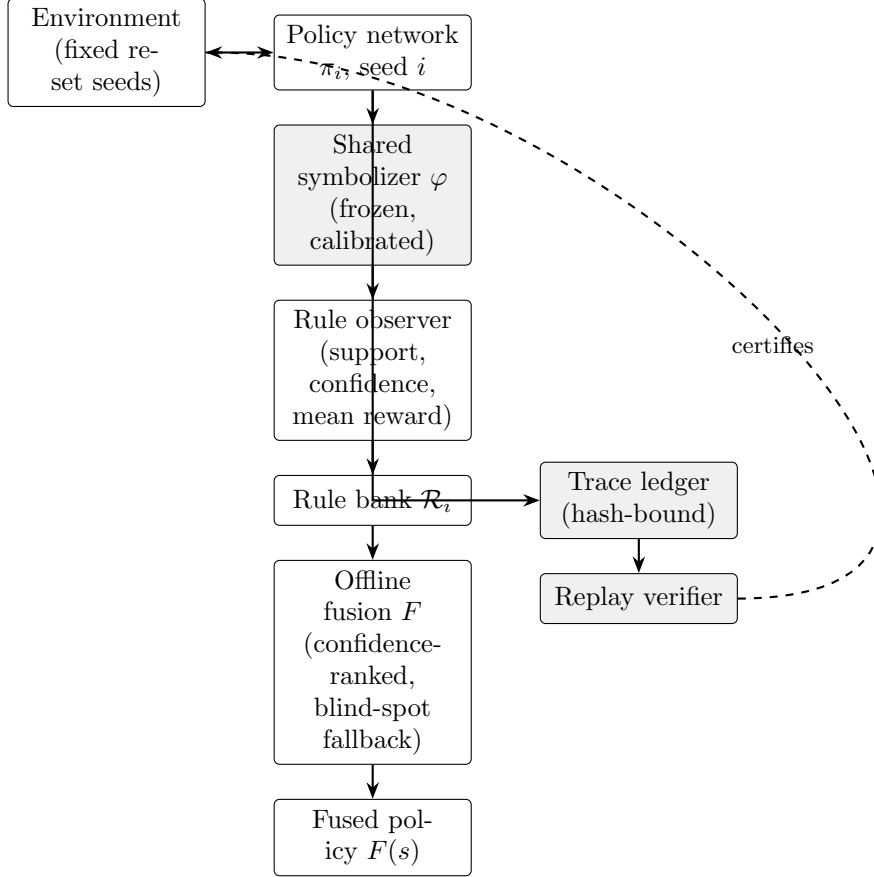


Figure 1: The auditable rule-induction protocol. A frozen shared symbolizer makes rule banks comparable across seeds; a hash-bound ledger and replay verifier certify fidelity; offline fusion composes rule banks into a single auditable policy. Dashed edge: the replay verifier re-executes the recorded actions in the environment under the recorded reset seeds.

canonical payloads plus exact environment replay, with the negative findings that human-readable serialization inflates size and that grammar-style codecs can fail to compress traces. These are practical reproducibility findings, not new cryptography.

4 Method

4.1 Shared state symbolizer

All agents operate on the same continuous state space, but independently trained networks may rely on different parts of it. To make descriptions comparable, we fix *one* symbolizer before any agent is trained and freeze it. For each state dimension $j \in \{1, \dots, d\}$, pooled empirical quantiles are computed from separate random-policy roll-ins (24 episodes per seed), yielding three cut points per dimension and four bins. The symbolizer is

$$\varphi(s) = (\text{bin}_1(s_1), \dots, \text{bin}_d(s_d)) \in \{0, 1, 2, 3\}^d, \quad (3)$$

so the shared vocabulary has 4^d symbols (256 symbols for CartPole-v1, 4096 for Acrobot-v1). The quantiles are computed once and never updated, so every seed and every arm of the experiment

sees the same abstraction. This design choice is coupled to everything downstream; we discuss its alternatives in Section 7.

The calibration roll-ins are generated by a uniformly random policy, not by any trained actor, and their reset seeds are disjoint from both the training reset-seed range ($10^4 \cdot \text{seed} + i$) and the evaluation reset-seed range ($2 \times 10^7 + i$); the calibration offset is 10^5 , so no calibration episode is ever a training or evaluation episode, although the same eight seed *identities* are reused for the roll-in streams. Quantile cut points are computed separately per environment, before any rule bank is constructed, and are never recomputed from held-out states. We treat the four-bin resolution as a design choice rather than an optimized representation: no bin-count search was performed, and the symbolizer is therefore a plausible confounder for every downstream result, as Section 7.5 states.

Definition 1 (State symbolization). *A state symbolizer φ maps a real state to a tuple of bin indices*

$$\varphi(s) = \left(\text{bucketize}_d(s^{(d)}; \theta_d) \right)_{d=1}^D, \quad (4)$$

where each θ_d holds the interior empirical-quantile edges of dimension d over $B = 4$ bins. The edges are fitted once from pooled random-policy calibration rollouts and then frozen, so that all training seeds and arms within a task share the same symbol alphabet.

Three consequences follow and are load-bearing. (i) The symbolizer is built from pooled calibration data; no seed fits its own discretizer. (ii) Shared labels are *not* shared semantics: when each agent fits its own state discretizer, identical bin labels need not denote identical raw-state regions—the largest observed edge difference under per-seed fitting was 0.157 observation units. (iii) Symbol overlap is therefore *not* behavioural equivalence. The fix that makes cross-seed comparison possible is the single frozen symbolizer above.

4.2 Rule bank and admission

A *rule* is a pair $(\mathbf{c}, a)$ where $\mathbf{c} = \varphi(s)$ is a condition (a discrete state symbol) and $a \in \mathcal{A}$ is an action. From the logged trace of agent i , the observer computes, for every condition–action pair,

$$n_{\mathbf{c},a} = \#\{(s, a) \in \mathcal{T}_i : \varphi(s) = \mathbf{c}\}, \quad f_{\mathbf{c},a} = \frac{n_{\mathbf{c},a}}{\sum_{a'} n_{\mathbf{c},a'}}, \quad \mu_{\mathbf{c},a} = \frac{1}{n_{\mathbf{c},a}} \sum_{r \in \mathcal{T}_i(\mathbf{c},a)} r, \quad (5)$$

where $\mathcal{T}_i$ is the transition set of agent i and r is the immediate reward. The condition–action pair is admitted into the rule bank $\mathcal{R}_i$ if and only if

$$\text{Adm}(\mathbf{c}, a) \equiv (n_{\mathbf{c},a} \geq \tau_s) \wedge (f_{\mathbf{c},a} \geq \tau_c), \quad \tau_s = 8, \tau_c = 0.70. \quad (6)$$

Definition 2 (Admitted state–action rule). *Let $N(c, a)$ be the count of observed transitions whose symbol is $c = \varphi(s)$ and action is a , and $N(c) = \sum_a N(c, a)$. The induced action for condition c is $a^*(c) = \arg \max_a N(c, a)$ (ties to the smallest index), with support $\kappa_s(c) = N(c, a^*)$ and confidence*

$$\kappa(c) = \frac{N(c, a^*(c))}{N(c)}, \quad \text{rule } c \rightarrow a^*(c) \text{ is admitted iff } \kappa_s(c) \geq \kappa_s^{\min} \wedge \kappa(c) \geq \tau. \quad (7)$$

The admitted rules form a rule bank $\mathcal{R}$. Each rule stores the multiset of source step identifiers that produced it, and the digest of that multiset.

Equation (7) uses a single normalization for confidence; support and confidence are reported separately from coverage, so that a precise-but-rare rule is never described as broadly applicable. The thresholds are not arbitrary. A high-confidence variant ($\tau_c = 0.90$) of the *online* rule-constraint pilot admitted so few rules that fusion deferred to the fallback on 100% of held-out decisions, and strict admission produced a 100% blind-spot collapse (Section 5.4); at $\tau_c = 0.70$ with $\tau_s = 8$ and an explicit blind-spot fallback, coverage failure becomes observable rather than silent. High confidence is therefore neither safety nor generalizability: rule admission is a *passive* observation step and must never re-enter the training loop.

We call $f_{c,a}$ the *confidence* of the rule. The admission predicate is a deterministic reduction of the trace: it compresses logged statistics without sampling, so its output is exactly reproducible from the same trace.

4.3 Audit ledger and replay verification

Fidelity is certified by two independent mechanisms.

Environment replay. For every recorded transition, the verifier re-executes the environment under the recorded reset seed with the recorded action and compares the resulting state, reward, termination, and truncation flags against the recorded values:

$$\text{Replay}((s, a, r, s', \text{term}, \text{trunc}); k) \equiv [\text{Env}(s, a; k) = (s', r, \text{term}, \text{trunc})], \quad (8)$$

where $\text{Env}(\cdot; k)$ is the deterministic environment transition under reset seed k and equality is exact float32 equality. Replay is a transport-level check: it certifies that the logged trace is the executed trace, and it says nothing about whether the policy is good.

Hash-bound ledger. Every training event (environment transition and optimizer update) is serialized into a canonical payload and appended to a hash chain

$$h_0 = H(\text{init}), \quad h_t = H(h_{t-1} \parallel \text{payload}_t), \quad (9)$$

with SHA-256. The chain is recomputed from the raw payloads and compared against the recorded chain; a mismatch at any position invalidates the entire run. An update ledger records, for each optimizer step, the checkpoint hash before and after the step, so the chain certifies the sequence of weight states as well as the sequence of transitions.

Three distinct evidence types. The audit ledger supplies three separable proofs that must not be merged into one “audit score”. (i) *Hash-chain integrity* certifies the record was not edited after the fact. (ii) *Environment replay* certifies that each recorded transition can be regenerated by the same environment under the same reset seed (a transport-level check). (iii) *Provenance resolution* certifies that a given decision can be parsed back to the rule, source bank, arbiter outcome, and fallback that produced it. The provenance predicate is

$$V_{\text{prov}}(t) = [\text{ref}_t \in \mathcal{R}_{i(t)} \wedge \text{act}(\text{ref}_t) = a_t \wedge \text{stat}(\text{ref}_t) = \text{stat}_t], \quad (10)$$

which an independent verifier evaluates for every decision in the fused trace.

The sense in which we use “certified”. Because these mechanisms are easy to over-read, we fix their meaning here. We use “certified” only in the operational sense just defined: a certification claim in this paper is relative to the recorded artifact, the environment version, the reset seeds, and the verifier implementation. It is not a certificate of the original training process, of the training

harness configuration, of optimizer or random-number state, or of the neural policy’s internal mechanism. Hash-chain integrity and exact replay establish that the stored record was not altered after the fact and that it can be regenerated under the recorded environment and reset seeds; they do not establish that the record was produced by the training procedure it claims, nor that the induced rules are a faithful semantic explanation of the network. Where the paper says a layer is “verified”, that is the claim being made and no more.

The hashable core. A subtle failure mode is a hash that covers non-core fields. In an early schema a sidecar counter and a wrapper label entered the update hash even though they did not affect optimisation, so two runs with identical optimizer behaviour hashed differently. The fix is a *normalized core-update hash* that excludes metadata (the previous hash, the record hash, and the grammar-generation annotation) and covers only optimisation-relevant tensors—step index, parameter-gradient hash, and pre/post checkpoint hashes. Equivalent runs then hash identically, and optimizer-equivalent arms (baseline vs. audit-only) are demonstrably byte-identical.

Definition 3 (Replay-and-provenance audit). *A fused trace $\tau = (\rho_1, \dots, \rho_T)$ passes the replay-and-provenance audit, written $\text{PA}(\tau) = \text{true}$, if and only if all three conditions hold: (i) environment replay passes for every step (the replayed transition equals the logged transition under the recorded reset seed, Eq. 8); (ii) provenance resolves for every decision (the cited rule reference belongs to the declared source bank, its action equals the emitted action, and its support and confidence meet the admission thresholds, Eq. 10); and (iii) the hash chain recomputes exactly (Eq. 9). The predicate is defined only over the audited traces under the specified experiment and artifact scope—the eight-seed, two-environment, shared-symbolizer study reported here—and asserts nothing about fused traces outside that scope, nor that the training-time logger was honest.*

4.4 Lossless trajectory codec

Beyond the rule bank, every trace is also compressed by a *lossless* codec so that a replayed record can be checked against its compressed form. The codec is a Re-Pair-style pair-substitution grammar with a bounded rule count (≤ 96 rules), and decoding is asserted to be exact (the decoded payload hashes to the input hash). Compression is measured as a two-part code length: the grammar description plus the encoded sequence, using Elias-gamma counts for terminals and a fixed-width symbol reference for non-terminals; the provenance bytes are reported separately. Writing the two-part description length as

$$L_2(\tau) = L(\mathcal{G}_\tau) + L(\tau \mid \mathcal{G}_\tau), \quad (11)$$

where $\mathcal{G}_\tau$ is the induced grammar and $L(\tau \mid \mathcal{G}_\tau)$ is the length of the trace encoded under it, we stress that $\mathcal{G}_\tau$ is a *lossless trace grammar*: it reconstructs the observed sequence exactly, but it is neither the state-action rule bank nor an explanation of the neural policy. We call L_2 a *two-part description length*, not a minimum description length: we do not claim optimality of the induced grammar, and we perform no model selection over a formally specified code family. The codec is *exact*, not a lossy summary: its role is fidelity, not storage. In practice it does *not* beat generic gzip on size (Section 6.10), which is itself a reported finding.

4.5 Training arms

Each seed trains three arms with identical hyperparameters and reset seeds:

- *Baseline*: unconstrained REINFORCE.

- *Audit-only observer*: identical to the baseline (same weights, same trace hash) but the rule observer runs alongside and emits a rule bank from the training trace. Its actor is byte-identical to the baseline actor, which we verify explicitly.
- *Constrained*: the action distribution is masked so that only actions admitted by rules extracted from the agent’s own earlier trace can be selected during training.

The constrained arm was part of the original design intuition (*let the rules steer the training*) and is retained in the experiment to quantify the cost of that intuition (Section 5.3).

4.6 Rule-level policy fusion

Given rule banks $\{\mathcal{R}_i\}_{i=1}^n$ from n seeds, the fused policy makes a decision at state s as follows (Figure 2). Let $\mathcal{C}(s) = \{r \in \bigcup_i \mathcal{R}_i : \mathbf{c}_r = \varphi(s)\}$ be the set of candidate rules whose condition matches the current state. If $\mathcal{C}(s) = \emptyset$, the state is a *blind spot* and the fused policy falls back to the action of the actor-logit ensemble (the argmax of the mean action logits over the n source actors). Otherwise, the fused action is the action of the candidate with the highest confidence; ties are broken by support, then by mean reward, then by source index. Formally,

$$F(s) = \begin{cases} \arg \max_a \max_{r \in \mathcal{C}(s)} w(r) \mathbf{1}[a_r = a], & \mathcal{C}(s) \neq \emptyset, \\ \pi_{\text{fb}}(s), & \mathcal{C}(s) = \emptyset, \end{cases} \quad (12)$$

where $w(r) = (f_r, n_r, \mu_r, -i)$ in lexicographic order and π_{fb} is the blind-spot fallback (the actor-logit ensemble argmax). Two per-decision quantities are recorded for every step:

$$\text{cov}(s) = \mathbf{1}[\mathcal{C}(s) \neq \emptyset], \quad \text{confl}(s) = \mathbf{1}[\#\{a_r : r \in \mathcal{C}(s)\} > 1]. \quad (13)$$

We call $\text{confl}(s) = 1$ a *conflict*: the seeds disagree about the action for this condition.

Decision classes. Every fused decision is classified into one of five classes, and the class is recorded alongside the action: *agreement* (all candidates agree), *single-source* (only one bank contributes), *conflict* (candidates disagree on the action), *blind spot* (no candidate; fallback used), and *fallback* (the residual single-actor action). These classes are not report formatting; they are the conditions under which the result is interpreted. Reporting a fused return without the coverage, conflict rate, blind-spot rate, fallback rate, source distribution, and arbitration outcome is incomplete.

4.7 Analysis instruments

Three instruments probe what the rules and the fused policy actually mean.

Behavioral agreement. For two actors π_i, π_j , sample N fresh states from each actor’s own occupancy and measure the proportion on which the two policies select the same action:

$$\bar{A}(\pi_i, \pi_j) = \frac{1}{N} \sum_{s \in \mathcal{S}_{ij}} \mathbf{1}[\pi_i(s) = \pi_j(s)], \quad (14)$$

where $\mathcal{S}_{ij}$ is the pooled fresh-state sample. Because the states are drawn by occupancy rather than uniformly, we also report an episode-cluster-balanced estimate and a cluster-balanced bootstrap interval.

Figure 2: Rule-level policy fusion with provenance record.

Input: rule banks $\mathcal{R}_1, \dots, \mathcal{R}_n$; symbolizer φ ; fallback actor π_{fb} .

For each decision at state s :

1. $\mathbf{c} \leftarrow \varphi(s)$
2. $\mathcal{C} \leftarrow \{r \in \bigcup_i \mathcal{R}_i : \mathbf{c}_r = \mathbf{c}\}$ (candidate rules)
3. **if** $\mathcal{C} = \emptyset$ **then return** $\pi_{\text{fb}}(s)$ and mark a *blind spot*
4. **if** all $r \in \mathcal{C}$ share one action **then** choose it and mark *agreement* (or *single-source* if $|\{i : r \in \mathcal{R}_i\}| = 1$)
5. **else** choose $r^* \leftarrow \arg \max_{r \in \mathcal{C}} (f_r, n_r, \mu_r, -i)$, attach its source bank and mark *conflict*
6. append (c, a, ν) to the decision log, update the chain digest, and return (a, ν)

Rule-set overlap. Rule banks are compared with the Jaccard similarity of their exact state-action sets,

$$J(\mathcal{R}_i, \mathcal{R}_j) = \frac{|\mathcal{R}_i \cap \mathcal{R}_j|}{|\mathcal{R}_i \cup \mathcal{R}_j|}, \quad (15)$$

together with the number of shared conditions whose recommended actions conflict.

Fitted-Q GPI baseline. To place the fusion comparison on a principled footing, we fit, for each frozen source actor π_i , an action-value regressor by fitted Q evaluation (FQE) [9, 10] over a pooled replay buffer:

$$\hat{Q}^{(k+1)}(s, a) = r + \gamma \hat{Q}^{(k)}(s', \arg \max_{a'} \pi_i(s')), \quad (16)$$

minimized in the least-squares sense over the replay transitions, with a reservoir cap of 25,000 transitions per seed and $\gamma = 0.99$. The baseline policy is the generalized policy improvement (GPI) selection [7]

$$\pi_{\text{GPI}}(s) = \arg \max_a \max_i \hat{Q}_i(s, a). \quad (17)$$

We emphasize that this is an *approximate* GPI: the classical GPI bound requires bounded value estimation error, which we do not claim to control; the fitted critics are diagnostics, and their failure is informative but not evidence about exact GPI.

4.8 Matched-support causal symbol intervention

The communication experiment (Section 5.7) is kept methodologically separate from rule fusion. In the redesigned *private-target* game a sender observes a private label and a receiver must act on the message alone; reward is given only for correct identification, so ignoring the message cannot be optimal. We measure the causal effect of a learned symbol with a *matched-support intervention*: take states where the sender emits a particular message, replace the message with a shuffled one (a permutation null), and measure the change in the receiver’s action probability. The intervention is matched so that only the message content changes. Under the controlled intervention protocol, which holds the receiver state and task context fixed to the extent implemented by the environment, we estimate the effect of replacing the discrete message on the receiver’s action probability. We distinguish observational association from intervention-based effects: in line with causal approaches to explaining RL behavior [17], the communication study replaces the candidate symbol while holding the task state and receiver context fixed as defined by the environment, which identifies only the local intervention effect in the designed game. It does not establish that the learned symbol has a human-interpretable meaning. A staged acceptance gate governs every causal claim:

diversity $\rightarrow$ positive control $\rightarrow$ necessity $\rightarrow$ generalisation $\rightarrow$ causality $\rightarrow$ channel exclusivity, each with a pre-registered verdict. The result is bounded to this designed task and depends on the stated common-support and no-private-channel checks: it is *not* evidence about state-action rule fusion, and it does *not* generalize to arbitrary neural policies.

4.9 Statistical conventions

All composition evaluations use 100 held-out episodes per policy, with reset seeds $20,000,000 + i$ for $i = 0, \dots, 99$; every policy sees the same episodes, so policy comparisons are paired at the episode level. Paired differences are summarized by the mean of the per-episode difference, with a 95% percentile bootstrap interval over episodes (5,000 resamples). Standard deviations are cross-episode sample standard deviations; we do not claim cross-seed variance estimates, because the seed pool ($n = 8$) is too small for that purpose. Where a quantity is deterministic (replay equality, hash validity), we state that variance is structurally zero.

5 From Intuition to Protocol: a Failure-Driven Design History

This section documents, in order, the failures that shaped the protocol. Each entry states the failure, its cause, and the change it forced. The sequence is part of the scientific record: it explains why the protocol looks the way it does, and it is evidence about which intuitions are unsafe.

5.1 Compression, replay, and explanation are different claims

The first intuition is that a recorded trace *is* the explanation: “here is exactly what the agent did.” A trace is faithful, but it is not a description. A full trace of a single seed on CartPole-v1 contains $\approx 97,646$ transitions (and up to 300,000 records per arm across the population); a human cannot inspect it, two traces cannot be compared except by exact matching, and a trace says nothing about states the agent never visited. The lesson, formalized as Section 4.3, is that traces belong to the fidelity layer, and a separate compression step is needed for the description layer. We therefore treat *trace* and *rule bank* as distinct artifacts with distinct evidentiary roles.

5.2 First-generation prototype failures

An initial prototype attempted to compress behavior with sequence grammar induction. Three defects were found, each independently fatal for the intended claim:

1. *Lossy grammar.* The prototype’s sequence-compression inducer was a simplified variant of SEQUITUR [18] and was not equivalent to the published algorithm. The induced grammar could not be proven to reproduce the source sequence, so any rule extracted from it was not anchored to the trace. Cause: reimplementing a linear-time grammar algorithm from memory without a conformance test. Fix: discard grammar-based compression as an evidence path; compress only *statistics* (support, confidence, mean reward) whose computation is a deterministic reduction of the trace.
2. *Conflation of one-step and window rules.* The observer initially treated “the agent chose *a* in condition *c*” and “the agent repeated a pattern across several steps” as the same kind of rule. They are different quantities: the first is a per-decision statistic, the second a sequence property. Mixing them produced rules with no well-defined estimand. Cause: one object

was used for two jobs. Fix: the protocol admits only per-decision condition–action rules (5); sequence structure is outside the evidence budget of this paper.

3. “Replay” meant different things. An early verifier checked that recorded actions, when replayed, reproduced *some* transition statistics; it did not check exact per-transition equality. Cause: the verifier tested aggregate health, not fidelity. Fix: exact float32 equality per transition (8), replayed from the recorded reset seed.

5.3 Rule-constrained training interferes with behavior

The original design intuition was that induced rules should actively guide training: mask out actions that violate the agent’s own rules. The implemented mask is state-conditioned and hard. On a state symbol that a rule covers, every other action is set to $-\infty$, so the sampled action is the rule’s action with probability one; on an uncovered symbol the action set is untouched. This supersedes an earlier *global* action-pruning variant that could remove an action permanently merely because it was absent from frequent rules.

A two-seed pilot ($\tau_c = 0.90$) was the first negative result, and it was sign-inconsistent rather than uniformly catastrophic: on seed 11 the constrained actor’s held-out return fell from 500 to 121.3 (-378.7), while on seed 29 it *rose* from 131.0 to 162.5 ($+31.5$). The induced bank covered only 0.41% of held-out steps on seed 11 and 6.98% on seed 29, so the intervention was close to inert on both. A high-confidence threshold does not make a constraint safe; it makes it rarely applied.

The eight-seed study ($\tau_c = 0.70$) confirms the heterogeneity (Table 2, Figure 3): on CartPole-v1, constraining training improved some seeds (29: 131.0 $\rightarrow$ 293.5; 71: 130.2 $\rightarrow$ 387.2; 43: 153.75 $\rightarrow$ 500.0) and damaged others (11: 500.0 $\rightarrow$ 182.3; 149: 500.0 $\rightarrow$ 96.1; 307: 500.0 $\rightarrow$ 161.2). On Acrobot-v1 every seed’s held-out return is -500 in both arms; that value is the step limit, not a measured difference, so the task contributes a null result rather than evidence of damage.

The cause is structural, and it is not a coverage failure. The threshold-relaxation experiment was designed to test the hypothesis that low coverage caused the damage, and it refutes that hypothesis: across the eight seeds the induced bank’s held-out coverage *rises* in 8/8 seeds (mean 0.477 $\rightarrow$ 0.761) while action agreement on covered states *falls* in 8/8 seeds (mean 0.887 $\rightarrow$ 0.676; Figure 3). What contracts is the policy’s own state occupancy, not the reach of the bank. On the training trace, where the learner does its exploring, the effective number of state symbols visited, 2^H , falls in 7 of 8 seeds (seed 11: 79.3 $\rightarrow$ 58.4; 149: 91.1 $\rightarrow$ 45.4; 307: 70.5 $\rightarrow$ 37.5; 43: 83.8 $\rightarrow$ 63.5; the exception is seed 211, 53.1 $\rightarrow$ 60.4). The same statistic is mixed on held-out evaluation, where it contracts in only 4 of 8 seeds: a policy that fails early visits more of the state space before it dies, so held-out occupancy conflates where the policy lives with how long it survives.

Two mechanisms are built into the implementation and together remove the learning signal the constraint is meant to shape (Figure 4). We establish that both are present; we do not apportion the damage between them. First, the mask deletes its own learning signal: when one action is left at probability one its log-probability is exactly zero, so a masked step contributes exactly zero to the REINFORCE objective $-\log \pi(a \mid s) G$. Between 36.7% and 74.4% of training steps were masked in our runs, so up to three quarters of the gradient budget was discarded at precisely the states the policy visited most often. This half is arithmetic rather than empirical: no experiment can strengthen it. Second, the observer re-enters the loop it is meant to observe. The rule bank is re-induced from the constrained trajectory every 50 episodes, and that trajectory contains only the actions the mask permitted, so the mask’s own choices become the evidence that re-admits the same rules. Measured directly, 68% of the observations the final bank rests on were generated by the mask itself (range 45%–85% across seeds). The observer’s counts are cumulative and are

Table 2: Per-seed held-out returns on CartPole-v1 (20 held-out episodes per seed, reset seeds $10^4 \cdot \text{seed} + 9 \times 10^5 + i$; each seed is evaluated on its own reset-seed range, so these rows are not the matched 100-episode comparison set of Table 7): baseline actor versus the rule-constrained actor. The constraint is a training-time intervention; its effect is heterogeneous and is negative for every seed that reaches the ceiling without it. On Acrobot-v1 every seed returns -500 in both arms.

Seed	Baseline	Constrained	Change
11	500.0	182.3	-317.7
29	131.0	293.5	$+162.5$
43	153.75	500.0	$+346.25$
71	130.15	387.15	$+257.0$
101	201.3	149.15	-52.15
149	500.0	96.1	-403.9
211	500.0	403.2	-96.8
307	500.0	161.2	-338.8

never reset, so a covered rule cannot be falsified by any later evidence: across all eight seeds and six re-induction generations we observe 0 retractions of an admitted rule and 0 changes to a rule’s forced action. The loop is self-confirming in the precise sense that the data which would falsify a rule are never collected.

Two measurements, two different questions. The rule-constrained arm is evaluated with the rule action substituted whenever a rule fires, which is 61%–95% of held-out steps. The return it reports is therefore that of the *hybrid* system—the learned policy and the frozen bank together, running as the implementation runs them—not of the network alone. We report both quantities, because they answer different questions and neither supersedes the other: the hybrid score asks what this intervention delivers when the bank is allowed to speak, and the network’s own score asks what the learner retained. To obtain the second we re-evaluated the same saved checkpoints on the same held-out reset seeds with the substitution disabled, following the protocol prior work used to study the removal of a trained mask [2]. The two readings disagree in an informative way (Table 3). As a hybrid the intervention costs 55.45 on average (271.6 against 327.0 for the unconstrained actors); stripped of the bank, the network is further below those actors (225.13), and four seeds lose more than 120 points. The difference between the two readings is the bank’s own contribution, which is positive on average ($+46.44$) and decisive on seed 149, where a network that alone scores 486.4 is scored 96.1 once the bank overrides it. Quoting only the hybrid number would understate the cost to the learned policy; quoting only the network’s own number would imply the bank is uniformly harmful, which it is not.

The same rule machinery that *composes* well offline (Section 6.5) is not dependable as a training-time intervention. Consequence: the protocol separates rule induction (audit-only observer) from rule composition (offline fusion); the constrained arm is retained as a quantified negative result that is heterogeneous and sign-inconsistent across seeds, not as evidence that induced rules are generally harmful.

5.4 Confidence thresholds and the 100% blind-spot collapse

Rule admission requires a confidence threshold. A high-confidence variant ($\tau_c = 0.90$) produced fusion policies that deferred to the fallback actor on 100% of held-out decisions: the strict threshold admitted so few rules that no condition was covered, and the “fusion” was the fallback in disguise.

Table 3: Two held-out readings of the rule-constrained arm on CartPole-v1, one per column group. “Hybrid” is what the arm reports: the composed system, bank included. “Network only” re-evaluates each constrained arm’s saved checkpoint with the rule substitution disabled, so it is the learned policy’s own return. “Unconstrained” is the baseline actor on the same reset seeds. The two readings answer different questions and are not alternatives to one another. A positive substitution gain means the frozen rule bank scored better than the network it was induced from.

Seed	Hybrid	Network only	Unconstrained	Substitution gain
11	182.3	127.8	500.0	+54.5
29	293.5	365.1	131.0	−71.6
43	500.0	217.9	153.8	+282.1
71	387.1	237.6	130.2	+149.6
101	149.2	80.0	201.3	+69.2
149	96.1	486.4	500.0	−390.2
211	403.2	120.8	500.0	+282.4
307	161.2	165.4	500.0	−4.2
Mean	271.6	225.1	327.0	+46.4

Cause: confidence is a conditional statistic; raising it shrinks support monotonically, and at the extreme the rule bank no longer covers the state space. The threshold was therefore set to $\tau_c = 0.70$ with $\tau_s = 8$ and an explicit blind-spot fallback (12) so that coverage failure is observable rather than silent. The episode is a general lesson: *coverage is not correctness*. Confidence, support, coverage, action agreement, and return are five different quantities; a high coverage or confidence threshold is a statistical choice, never a semantic validation of the rule. This is a two-setting exploratory choice, not a systematic threshold search; we report the boundary in Section 6.

5.5 Serialization and ledger-schema failures

Two failures concerned the audit artifacts themselves. First, a human-readable (pretty-printed) serialization of the trace inflated its size by roughly 1.8×; after switching to a compact canonical encoding, the compressed trace ledger was still 8.4% larger (CartPole-v1) and 9.9% larger (Acrobot-v1) than the gzip-compressed raw trace, because the grammar-style codec adds structure without removing bytes. The codec is nevertheless exact: the decoded payload hashes to the input hash. We report these costs as part of practical reproducibility. Second, the update-ledger schema initially allowed a sidecar field (a grammar-generation annotation) to vary without changing the optimizer record, so two runs with identical optimizer behavior could hash differently. Cause: the hash covered non-core fields. Fix: the ledger hashes only a normalized core record (step index, parameter gradient hash, pre/post checkpoint hashes), making equivalent runs hash identically.

The same lesson appears in the communication task below (Section 5.7): a sidecar field (a grammar-generation annotation) once entered the update hash even though it did not affect optimisation, and the fix was to hash only a normalized core record. An audit instrument must audit itself.

5.6 Arbitration sensitivity and the random-selection anomaly

The confidence-ranked arbiter was chosen as the default resolver because it is the natural generalization of the per-rule confidence statistic. The results (Section 6.6) show it is the wrong default on

CartPole-v1 rule-constrained arm, 8 seeds: coverage rises, agreement falls

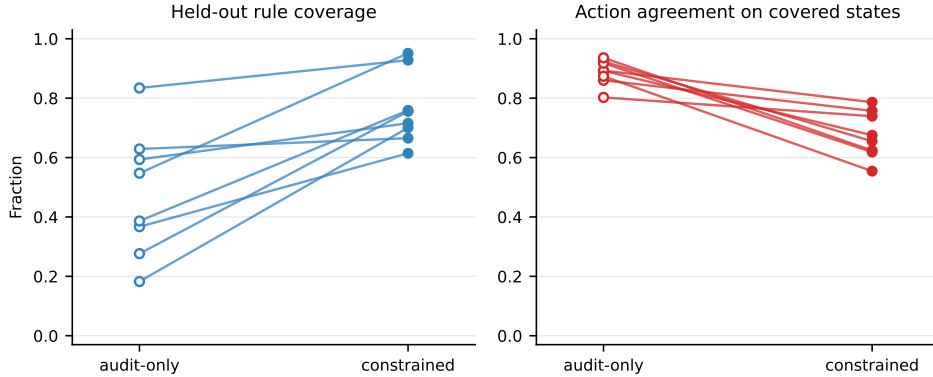


Figure 3: CartPole-v1 rule-constrained arm, eight seeds, $\tau_c = 0.70$. Left: held-out coverage of the induced rule bank. Right: action agreement between the bank and the learned policy on covered states. Each line is one seed; hollow markers are the audit-only arm and filled markers the constrained arm. Every arm is measured on its own state occupancy, so this is a same-seed pairing, not a matched-state comparison. Coverage rises in 8/8 seeds and agreement falls in 8/8 seeds: the mask does not reduce the reach of the rule bank, it removes the policy’s ability to disagree with it.

Acrobot-v1: uniformly random arbitration among candidate rules beats every deterministic variant tested, including the deployed one. We report this as an *open mechanism* rather than a method. Treated as a design event, it demotes the arbiter from a trusted component to a diagnostic: the fused policy is reported together with its full decision mix, and the random-selection ablation is part of the result, not a footnote.

5.7 Communication-task redesign

The causal-symbol evidence rests on a designed coordination game that had to be rebuilt twice. The original *parity* game (sender sees a digit, receiver acts on the message, reward on parity match) collapsed to a degenerate equilibrium: both policies settled on the constant action C/C, the unique joint-reward optimum under both parities, so the channel carried nothing. Under a parity-matching reward, constant actions are a Nash equilibrium and no optimizer can escape an objective whose global optimum ignores the channel—the failure was in *task design*, not learning.

A second, abandoned early design (the private-target variant) failed for a different reason: the source review found the original coordination payoff made the same joint action optimal for both parities; a blind control masked only the label while the image encoder retained parity information; a reserved index lay outside the embedding range; the codebook was bypassed by a straight-through continuous path; and an opponent-prediction branch was degenerately solvable by copying. The fix was a parity-critical payoff with opposed optima, joint masking of label and image pathways, corrected index handling, straight-through codebook wiring with perplexity reporting, and the staged gate above.

Even the redesigned game initially appeared to fail: after training, receiver accuracy was 0.500 (chance) and the estimated causal influence $\Delta P(C) = 0.00088$ (95% CI $[-0.00136, 0.00320]$) was indistinguishable from zero. The cause was *optimization exposure*, not architecture: the run had performed only 79 optimizer updates; reducing the batch size from 128 to 8 raised the count to

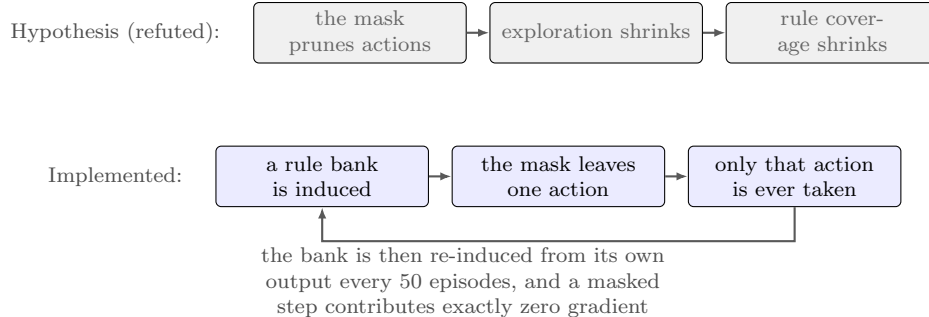


Figure 4: Mechanism of the training-time constraint, as implemented. The top row is the hypothesis that the threshold-relaxation experiment refutes. The bottom row is the loop the code implements: a hard one-hot mask, whose log-probability at a masked step is exactly zero and which therefore contributes nothing to the policy gradient, together with a rule bank re-induced from the masked trajectory every 50 episodes, which makes the mask’s own output the evidence for its own rules.

1,250, and a run with 12,500 updates succeeded decisively. Two analysis methods also produced impressive but empty numbers: a “symbol lookup” analysis was tautological (the action decoder read the first token directly), and a whole-sequence mutual-information estimate saturated at 1 bit because sparse, high-cardinality messages make plug-in MI vacuous. Both were replaced by a per-position MI against a permutation null. The only surviving causal claim comes from the redesigned task and is bounded to it.

6 Experiments and Results

6.1 Setup

Experiments use CartPole-v1 (4 state dimensions, 2 actions) and Acrobot-v1 (6 dimensions, 3 actions) from the Gymnasium suite [19]. Eight seeds {11, 29, 43, 71, 101, 149, 211, 307} are used; each seed trains 300 episodes per arm with REINFORCE (2) and a shared frozen symbolizer (3) calibrated from 24 random-policy episodes per seed. Rules are admitted with $(\tau_s, \tau_c) = (8, 0.70)$. Composition evaluates 100 held-out episodes per policy on the common reset seeds. All reported traces were post-hoc audited: hash chains, update ledgers, baseline/observer equivalence, and environment replay were re-verified after the runs completed.

Because deep RL results can vary substantially with random seeds, environment nondeterminism, evaluation protocol, and reporting choices [20], we report seed identities, matched reset seeds, evaluation distributions, and uncertainty intervals separately rather than folding them into a single number: the seed pool is small ($n = 8$), the reported intervals are over episodes and not over seeds, and the comparator for the headline comparison is selected post hoc (Section 7.5).

6.2 Registered predictions

We fix predictions before reporting any result, so that they can be contradicted. They are *unregistered* (no external preregistration record exists), so they are treated as post hoc when assessing claim strength. Each receives a status in Section 6.11.

[‡]P6 holds on CartPole-v1 only under the narrow reading of “best single actor” as the actor selected by training return. Against the best *held-out* actor on the same matched episodes it fails by 294.04 return points (Table 8), so we record the prediction as comparator-dependent rather than

Table 4: Predictions fixed before analysis. Layer: operational-health (OH), diagnostic (D), identification-bearing (ID). Status assigned in Section 6.11.

ID	Prediction	Layer	Status
P1	Hash-chain, ledger and replay checks pass on all recorded runs.	OH	supported
P2	The lossless codec reconstructs every trace prefix exactly.	OH	supported
P3	Baseline and audit-only arms are training-equivalent.	OH	supported
P4	Cross-seed rule overlap is substantially above zero under a shared symbolizer.	D	supported
P5	Two actors with high rule overlap also agree on most fresh held-out states.	ID	contradicted
P6	Fused mean return exceeds the best single source actor on both tasks.	ID	comparator-dependent [‡]
P7	Blind spots, not conflicts, are the main obstacle to fusion.	ID	contradicted
P8	The deployed confidence-ranked arbiter is at least as good as alternative selectors.	ID	contradicted
P9	A learned rule bank predicts an independent holdout above an action-frequency baseline, with coverage, agreement, and confidence reported separately.	ID	contradicted (at floor)

supported. On Acrobot-v1 all eight source actors tie at the floor, so the two readings coincide there and the prediction holds.

P9 (the action-frequency baseline prediction) is reported here rather than deferred, because coverage/agreement statistics are otherwise undefined relative to a floor: a rule bank that agrees with the actor on 51% of states is only informative once compared with the majority-action frequency of the same states. On the same 10,000 fresh states used for the agreement study (Section 6.7), the occupancy-weighted majority-action baseline is 51.32% (majority action 2528/5000 on the seed-11 occupancy and 2604/5000 on the seed-29 occupancy), against an observed cross-actor agreement of 51.38%. The agreement therefore sits essentially at the action-frequency floor (margin +0.06 points), so the rule bank does *not* predict the holdout above the baseline; the prediction is reported as contradicted.

6.3 Audit integrity

Table 5 reports the fidelity layer. Every hash chain is valid, every update ledger verifies, baseline and audit-only actors are byte-identical, and every recorded transition replays exactly under its reset seed. The lossless codec round-trips exactly in both environments. These are operational-health results: they certify the artifacts, and they do not by themselves support any claim about policy quality.

Tamper probe. Hash validity is a claim about tamper detection, so we test it directly. We modified the reward field in the first record of a *copy* of the audited CartPole merged trace—the 100,000-decision, $\tau_c = 0.70$ fused trace whose 68,409 chosen rule references resolve to source grammars and whose 41,728 fallback references resolve to the selected seed-11 actor—and deliberately

Table 5: Audit integrity (operational-health layer). Values are deterministic; variance is structurally zero.

	CartPole-v1	Acrobot-v1
Trace records (all three arms)	1,725,655	2,544,513
Optimizer update records	7,200	7,200
Environment replay checks (post-hoc)	1,122,403	1,742,570
Constraint rule-reference verifications	290,052	697,817
Hash chains valid	yes	yes
Update ledgers valid	yes	yes
Baseline vs. audit-only actor identical	yes	yes
Codec round-trip exact	yes	yes
Raw trace bytes / gzip	61.3 MB / 12.3 MB	57.6 MB / 12.5 MB
Codec archive (gzip)	13.4 MB	13.7 MB

Table 6: Rule structure across seeds. Pairwise statistics use all 28 seed pairs.

	CartPole-v1	Acrobot-v1
Rule count per seed (range)	100–145	77–303
Conditions shared by ≥ 2 seeds	165	445
All-seed intersection (conditions)	38	2
same-action core	38	0
conflicting	0	2
Mean pairwise rule Jaccard (15)	0.557	0.096
Shared conditions (seeds 11 vs. 29)	77	98
conflicting (seeds 11 vs. 29)	0	77

did *not* recompute the chain. The replay-and-provenance validator (Def. 3) rejected the copy at step 1 with a hash mismatch, and the original trace was left unmodified. This is a tamper probe on the audited CartPole trace only; it is not evidence that every fused trace in every environment has been subjected to the same test.

6.4 Rule structure across seeds

Rule banks are seed-specific but structurally related (Table 6). On CartPole-v1, all eight seeds share a core of 38 conditions on which they *agree* (same action in every seed), and the mean pairwise Jaccard overlap of rule sets is 0.56; the two-seed comparison used throughout the report is 0.58. On Acrobot-v1 the picture inverts: the all-seed intersection contains only 2 conditions, and both are *conflicting*; the mean pairwise Jaccard is 0.096. Across the two environments, 77 of 98 shared conditions conflict in Acrobot-v1 versus 0 of 77 in CartPole-v1.

6.5 Composition on CartPole-v1

Table 7 and Figure 5 report the main composition result. On CartPole-v1, the rule-level fusion policy reaches a mean held-out return of 205.96 (median 181, cross-episode std 79.1) versus 163.65 for the single-actor comparator, a paired difference of +42.31 with 95% bootstrap interval [23.98, 61.93].

Table 7: Primary composition results. Means and paired differences over 100 common held-out episodes; 95% bootstrap intervals in brackets. “Best single actor” is the source actor with the highest training return mean (seed 43 on CartPole-v1, seed 211 on Acrobot-v1); the selection is post hoc and is analyzed in Table 8.

Policy	CartPole-v1	Acrobot-v1
Best single actor	163.65	−500.0
Actor-logit ensemble	500.0 (ceiling)	−500.0
Grammar fusion F (12)	205.96	−443.37
Fitted-Q GPI (17)	9.34	−499.57
Paired Δ fusion vs. best single	+42.31 [23.98, 61.93]	+56.63 [39.32, 75.15]
Paired Δ ensemble vs. best single	+336.35	0.0
Fusion coverage / conflicts / blind spots	98.3% / 75 / 356	95.7% / 39,602 / 1,909

Table 8: CartPole-v1 comparator sensitivity on the *matched* evaluation set (the same 100 common reset seeds used in Table 7). Every row is evaluated on the identical reset seeds, so the rows are directly comparable; the two single-actor rows differ only in the selection rule. The seed-43 row and the ensemble row reproduce the recorded values of Table 7 exactly, which validates the re-evaluation harness. Paired Δ is reported relative to rule fusion F (Eq. 12); it is undefined for the population summaries.

Comparator	Mean return	Paired Δ vs. fusion
Best <i>training</i> -return actor (seed 43)	163.65	+42.31 [23.98, 61.93]
Best <i>held-out</i> actor (seeds 11/149/211/307, tie)	500.0	−294.04 [−308.97, −278.16]
Mean over the eight actors	330.86	—
Median over the eight actors	358.81	—
Worst actor (seed 29)	131.11	+74.85
Actor-logit ensemble	500.0	−294.04
Rule fusion F (12)	205.96	—

The comparator is not a pre-declared baseline: it is the source actor with the highest *training* return mean (seed 43), which is a post-hoc choice, and the task exhibits ceiling effects—four of the eight actors reach the 500-step ceiling on their own held-out evaluations. We therefore report this comparison as *exploratory* rather than as a pre-declared win over the best baseline, and Table 8 reports how sensitive the headline is to the selection rule. That sensitivity is large and runs against us: on the same matched episodes, four of the eight individual actors reach the ceiling (500.0), the mean and median over the eight actors are 330.86 and 358.81, and fusion (205.96) is 294.04 points below the best held-out actor (−294.04, 95% interval [−308.97, −278.16]). Fusion exceeds only the comparator that the training-return rule happens to select, and the gain is therefore a statement about that selection rule, not about fusion being a strong single policy in this environment. Fusion decisions are 98.3% rule-covered (20,240 of 20,596 steps) with only 75 conflicting decisions (0.4%) and 356 blind spots (1.7%), which is consistent with the low-conflict rule structure of Table 6. The actor-logit ensemble saturates the environment ceiling (500.0, zero variance), which makes it a ceiling-level comparison rather than a differentiating one.

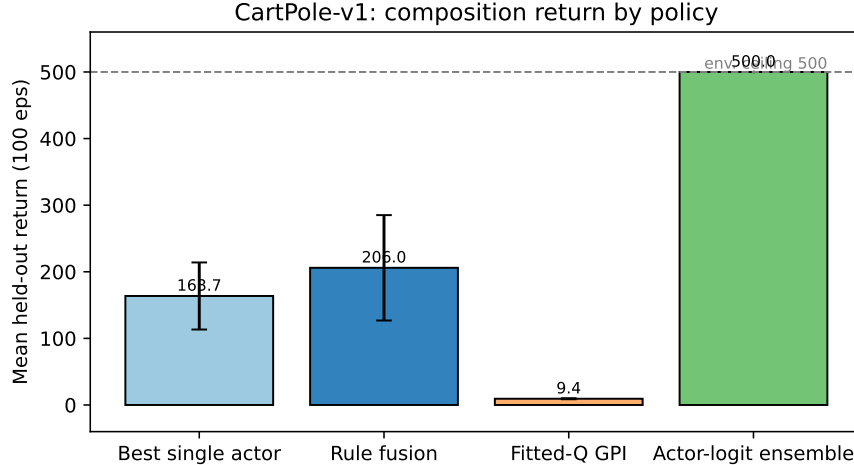


Figure 5: CartPole-v1 composition comparison. Error bars are 95% bootstrap intervals of the mean over the 100 held-out episodes. The dashed line marks the environment ceiling (500). The actor-logit ensemble and the FQE-GPI baseline bracket the fusion result from above and below; FQE-GPI is reported here and analyzed as a failed diagnostic in Section 6.9.

6.6 Acrobot-v1: the conflict regime and the failure of confidence arbitration

On Acrobot-v1, every source actor is at the floor (-500), yet the fusion policy is not: it averages -443.37 (median -500 , std 90.8), with 39 of 100 episodes terminating before the 500-step cap and a maximum of -126 . The paired difference versus the best single actor is $+56.63$ [39.32, 75.15]. However, the fusion decisions are 95.7% rule-covered and 89.2% *conflicting* (39,602 of 44,376 steps), so the fused policy is almost entirely an arbitration among disagreeing seeds.

The arbitration rule matters enormously, and not in the intended direction (Figure 6, Table 9). Uniformly random arbitration among candidate rules returns -187.02 on the primary evaluation (95% paired interval [231.38, 278.21] relative to confidence-ranked fusion) and is replicated across five independent random-number streams with means from -206.17 to -194.10 (overall -198.82). By contrast, confidence-ranked arbitration returns -443.37 ; mean-reward-first, support-first, defer-to-ensemble, and unanimous-only arbitration all return -500 ; and the single-source fallbacks return -499.5 and -500 . A matched control that ignores the rules entirely and samples uniformly random actions averages -498.71 across five replicates. Confidence-ranked arbitration is therefore not merely suboptimal on this task but *anti-selective under the evaluated conflict and occupancy regime*: choosing rules by confidence selected worse actions than choosing them at random.

A 200-state Monte-Carlo action-semantics probe (fresh states from deterministic actor rollouts) dissects the failure (Table 10). The fusion policy covers all 200 states, conflicts on 90.5% of them, and agrees with the best first action (counterfactual continuation return under the actor-logit ensemble) on only 22.5% of states; its action histogram also differs sharply from the Monte-Carlo-optimal histogram. That agreement rate is below the 26.7% expected if the fusion action histogram and the Monte-Carlo-best histogram were independent, so the confidence-ranked action is negatively associated with the estimated best first action *in this probe*. The mean margin between the best and second-best first actions is 108.24 return units, so the decisions are not near-ties. The confidence signal that arbitration relies on does not track the Monte-Carlo ranking of first actions in this environment.

Table 9: Acrobot-v1: arbitration ablations on the 100 matched held-out episodes. Paired intervals are bootstrap 95% intervals of the per-episode difference relative to the current confidence-ranked policy.

Arbitration rule	Mean return	Paired 95% interval vs. current
Confidence-first (current)	-443.37	—
Confidence only (no tie-breaks)	-444.49	[-3.16, 0.06]
Random candidate	-187.02	[231.38, 278.21]
Mean reward first	-500.0	[-75.47, -39.67]
Support first	-500.0	[-74.84, -39.68]
Defer conflicts to ensemble	-500.0	[-75.12, -40.08]
Unanimous only	-500.0	[-74.49, -39.36]
Best single fallback	-452.69	[-19.51, 0.50]
Single source 11 / 29	-499.48 / -500.0	—

Table 10: Acrobot-v1 Monte-Carlo action-semantics probe on 200 fresh states (state sample from new deterministic actor rollouts). “Best first action” is the action maximizing counterfactual continuation return when followed by the actor-logit ensemble.

Quantity	Value
Fusion coverage	100%
Fusion conflict rate	90.5%
Fusion agreement with any source- Q argmax	64.5%
Fusion agreement with ensemble-GPI argmax	0.0%
Fusion agreement with MC-best first action	22.5%
MC continuation return, first action 0 / 1 / 2	-132.9 / -202.5 / -157.1
MC-best first-action histogram (0/1/2)	87 / 46 / 67
Fusion action histogram (0/1/2)	0 / 129 / 71
Mean best-vs-second margin	108.24

6.7 Behavioral agreement: rule overlap does not transfer

Rule-set overlap and behavioral agreement measure different things, and the data show how different they are (Figure 7). On CartPole-v1, the rule banks of seeds 11 and 29 share a Jaccard overlap of 0.58 and agree on all 77 shared conditions, and the mean pairwise rule overlap is 0.56. Yet on 10,000 fresh states (5,000 from each actor’s own occupancy), the two policies select the same action on only 51.38% of states (episode-cluster-balanced estimate 51.36%; 95% bootstrap interval [50.48, 52.26]). Within each actor’s own occupancy the agreement is 50.6% and 52.2% respectively. Stratifying by pole angle shows the agreement is near chance for near-zero angles (50.2%) and only modestly above chance in the mid range (55.8%). The rules of these two agents share a Jaccard overlap of only 0.58, yet the agents behave almost independently on fresh states. Rule-set overlap is therefore not a proxy for behavioral agreement; it is an upper-layer statistic that must be validated against the behavior itself.

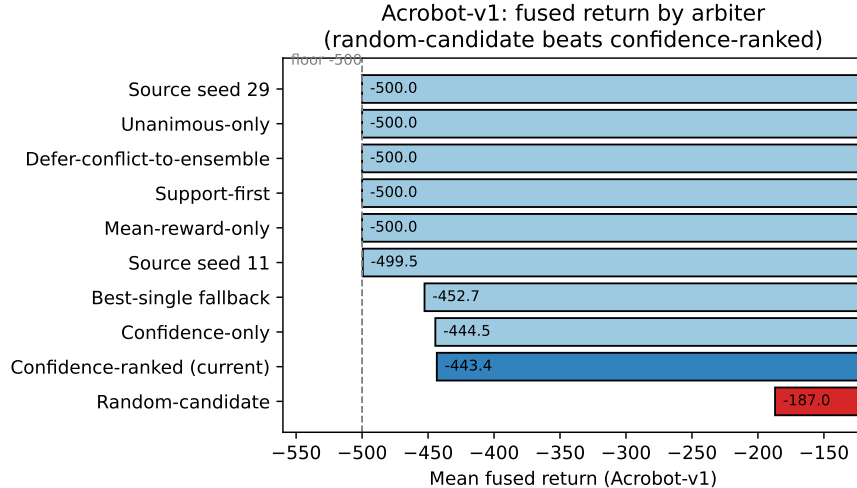


Figure 6: Acrobot-v1 arbitration ablations on matched episodes. Random arbitration among candidate rules (solid red bar) substantially outperforms confidence-ranked arbitration; the dashed line marks the five-stream random-arbitration mean. Uniform random actions (ignoring the rules) average -498.71 and are omitted for scale.

6.8 Causal symbol intervention as a separate sub-study

The communication experiment is reported apart from rule fusion because it answers a different question: whether a *learned discrete symbol* in a designed task causally influences a receiver’s decision, not whether state–action rule fusion composes policies. On the redesigned private-target game, six independent runs of 100,000 decisions each (600,000 total) give receiver accuracy 0.998923 with the true message, 0.503292 with a shuffled message, and 0.500 with no message. The matched-support intervention—replacing the message while holding the state fixed—changes the receiver’s action probability by $\bar{\Delta} = 0.998366$ on average (range 0.996986–0.999685 across runs), with common-support overlap 0.94308. In the seed-0 probe, per-position total-variation distance between even/odd token marginals is 0.0967 at position 0 and 0.9984 at position 1, confirming the information-bearing position identified by the per-position MI analysis.

Under the controlled intervention protocol, which holds the receiver state and task context fixed to the extent implemented by the environment, we estimate the effect of replacing the discrete message on the receiver’s action probability; the claim is internal to this designed task and depends on the stated common-support and no-private-channel checks. Its scope is deliberately narrow: it shows that *in this designed task* the learned discrete symbols causally influence the receiver’s decisions. It does *not* show that the symbols are human-interpretable, compositional, or “meaningful” in any richer sense, and it does not generalize to other RL agents. All causal claims in this paper rest on the redesigned task and the staged gate of Section 4.8; the original parity game is reported only as a negative result.

6.9 Fitted-Q GPI baseline failure

The fitted-Q GPI baseline (16–17) fails in both environments, and the failure mode is informative. On CartPole-v1 its mean return is 9.34 (paired difference -154.31 versus the best single actor), and the fitted critics are nearly flat: the critic values have cross-seed agreement of 0.84 on greedy actions, but their pooled Q -value distribution has a standard deviation of only 0.81 around a mean

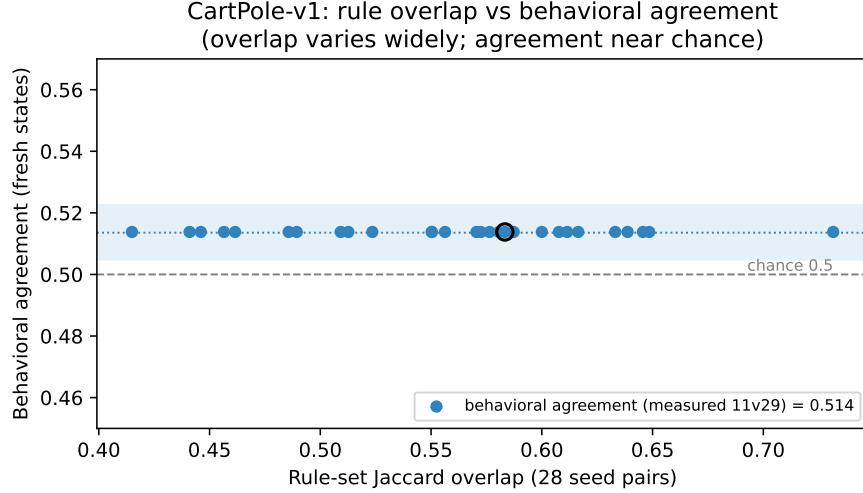


Figure 7: CartPole-v1: rule-set overlap (Jaccard, seeds 11 vs. 29 and all 28 pairs) versus behavioral agreement on 10,000 fresh states. Error bar: 95% bootstrap interval of the cluster-balanced agreement. The dashed line marks chance (0.5).

of 10.7, i.e., the critics vary little across states—a calibration/composition failure rather than a usable value signal. The FQE-GPI action histogram is 30,758:805 (nearly always action 0), and its agreement with the actor-logit ensemble on 31,563 held-out states is 35.3%. On Acrobot-v1 the collapse is total: FQE-GPI selects action 0 on all 50,000 held-out states, the critics are constant at -11.44 with standard deviation 0.009, greedy-action agreement between critics is 6.3%, and the held-out Bellman residual is 0.92 (MAE), which is far from the precision that the GPI bound would require. These are diagnostics, not discoveries: they show that fitted-Q composition over a 200k-transition reservoir is unreliable in these setups, and they caution against claiming that the rule-fusion result is “better than value-based composition” on the strength of this baseline alone.

6.10 Lossless coding results

The codec of Section 4.4 is exact by construction (the decoded payload hashes to the input hash). On CartPole-v1 it processes 9,178,724 tokens across 16 grammar productions; the raw trace is 61.3 MB, gzip compresses it to 12.3 MB, and the grammar-gzip archive is 13.4 MB. On Acrobot-v1 the corresponding figures are 8,549,496 tokens, 57.6 MB raw, 12.5 MB gzip, and 13.7 MB grammar-gzip. The two-part code length therefore *does not* beat generic compression: the grammar adds structure without removing bytes, and the grammar-gzip archive is 8.4% larger than gzip on CartPole-v1. The two-part code length $L_2(\tau)$ (Eq. 11), expressed as a ratio to the uncompressed raw trace, sits at 0.51–0.58 across the six 50-episode prefixes of a separate seed-11 CartPole probe; these prefix-level two-part code lengths come from that single seed-11 probe and not from the full eight-seed archive. The two-part code is shorter than the uncompressed reference, but the stored grammar archive is not smaller than gzip. The result is reported honestly as a fidelity tool, not a storage win: the codec earns its place by enabling exact replay-against-compressed-form checks, not by shrinking the log.

Table 11: Summary of measurement layers. Identification-bearing results are distinguished from diagnostics and operational-health checks.

Criterion	Layer	CartPole-v1	Acrobot-v1
Trace/hash/replay validity	operational health	pass	pass
Rule overlap across seeds	diagnostic	0.557	0.096
Fusion vs. best-training actor (paired Δ)	measured, capped	+42.31	+56.63
Fusion vs. best held-out actor (paired Δ)	measured, capped	−294.04	+56.63 [†]
Fusion mean return	measured	205.96	−443.37
Random arbitration vs. confidence	identification	n/a	−187.02 vs. −443.37
Behavioral agreement (fresh states)	identification	51.4% ($\approx$ chance)	n/a
FQE-GPI mean return	diagnostic	9.34	−499.57

6.11 Summary of verdicts

[†]On Acrobot-v1 all eight source actors return −500.0 on the matched set, so the best held-out actor and the post-hoc best-training actor coincide and both give +56.63; on CartPole-v1 they differ sharply, which is the point of the row.

Table 11 separates the layers. The identification-bearing findings are (i) the paired fusion advantage on CartPole-v1 against the post-hoc-selected comparator—reported together with its reversal against the best held-out actor, so that the row is not read as a general performance claim, (ii) the behavioral-agreement near-chance result, and (iii) the Acrobot arbitration reversal. All other rows are diagnostics or health checks.

7 Discussion

7.1 Why the mechanism may work, and where it breaks

The CartPole advantage of rule-level fusion has a simple account in the equations. The fused decision (12) selects the highest-confidence rule among the candidates (6); when conflicts are rare (0 of 77 shared conditions), the fused policy is essentially a union of the seeds’ agreement, and the fallback (12, second case) covers the residual 1.7% of states. The paired comparison controls for the reset-seed distribution, and the bootstrap interval excludes zero. But the same equations fail on Acrobot-v1 because the confidence statistic $f_{c,a}$ (5) is a *within-trace* conditional frequency, not a predictor of which action is best in a new state: when 89% of decisions conflict and the best first action (probe, Table 10) disagrees with the fused choice on 77.5% of states, the arbitration is selecting on a signal that does not rank actions by return. The reversal—random arbitration beating confidence arbitration—is consistent with the Monte-Carlo probe: in this Acrobot probe the confidence-ranked action agreed with the estimated best first action on only 22.5% of the 200 probe states, which is below the 26.7% expected if the fusion action histogram and the Monte-Carlo-best histogram were independent, so removing the signal helps. We treat this as an environment- and data-collection-specific diagnostic, not as evidence that confidence is generally anti-selective. We state it as an *open mechanism*: we can show the signal is uninformative here, but we cannot yet explain why it is *negatively* informative on Acrobot-v1.

7.2 Prediction outcomes

Two expectations held and two failed. Expected: (i) trace fidelity (replay, hashes) would verify cleanly, and (ii) rule overlap would be higher under a shared symbolizer than under per-seed

symbolizers (the design motivation). Unexpected: (iii) behavioral agreement would track rule overlap—it does not (51.4% vs. chance), and (iv) confidence-ranked arbitration would dominate random arbitration—it is reversed on Acrobot-v1. For (iii), an alternative explanation is that the shared symbolizer is too coarse: with only four bins per dimension, two policies can agree on rules for the symbol while disagreeing on nearly every state inside the symbol; the rules describe the *symbol*, and behavior lives on *states*. For (iv), an alternative explanation is that the trace’s confidence is dominated by the early, poorly-performing part of training in Acrobot-v1 (where all returns are -500 and mean reward is constant at -1.0 for 93% of rules), so confidence reflects state-visitation frequency rather than action quality. Both alternatives are testable and are the natural next experiments.

Occupancy action histograms as one mechanism for near-chance agreement. The behavioral-agreement result has a concrete mechanistic reading. On the 5,000 states drawn from seed 29’s held-out occupancy, seed 11 selects action 0 on 4994/5000 states, whereas seed 29 itself splits nearly evenly (2604 action 0 versus 2396 action 1); the asymmetry inverts on seed 11’s occupancy, where seed 29 selects action 1 on all 5000 states and seed 11 splits 2472/2528. Each actor is therefore close to deterministic inside the other’s occupancy, and the two opposite, near-uniform action preferences yield a pooled agreement near the binary chance baseline. We report this as a mechanism *consistent with* occupancy asymmetry as one explanation; we do not claim it is the unique cause, and the coarse four-bin symbolizer remains a candidate confounder.

7.3 What the protocol does *not* establish

The positive results are bounded, and the boundaries are the point. Specifically: (i) rule overlap does not imply behavioural agreement (Section 6.7); (ii) trace replay does not imply causal explanation, and re-executing recorded transitions recovers the record but not the optimizer state or random-number stream that produced it; (iii) provenance visibility does not imply recovery of the neural mechanism; (iv) the fitted-Q baseline failing its own gate does *not* imply that rule fusion is better than value-based composition—only that the comparator is unqualified here; (v) the random-arbitration anomaly does *not* imply that a random arbiter is the generally optimal resolver. A plain actor-logit ensemble also outperforms rule fusion in CartPole (500.00 vs 205.96) while being simpler, and so does the population it summarises: on the matched episodes the mean and median individual actor reach 330.86 and 358.81, and four of the eight actors reach the ceiling (Table 8). The objection is correct about performance and misses the point about auditability. The ensemble cannot say *which* knowledge produced a decision, cannot report that two sources disagreed, and cannot declare a blind spot. What fusion buys over the ensemble is not return but *inspectability per decision*; what the hash chain buys over a log file is not accuracy but *checkability*. Whether that tradeoff is worth it depends on the deployment; the claim is only that the tradeoff exists and can be measured honestly.

7.4 Theoretical and practical implications

Theoretical implication: under a shared, frozen symbolizer, rule-level composition inherits provable properties only for the statistics that define it (5–6); no guarantee about held-out behavior follows from rule overlap, and none should be claimed. The behavioral-agreement result is direct evidence for this boundary. Practical implication: rule fusion is usable as an *audit and composition* layer in low-conflict regimes, and is not a reliable training-time constraint (Section 5.3): that intervention is sign-inconsistent across seeds and damages the seeds that already reach the ceiling. We therefore

make no claim that fusion trains better policies, and we note that the constrained arm’s effect is heterogeneous rather than uniformly negative.

Auditability as a vector, not a score. The six predicates of Section 1.1 should be reported as a vector, never averaged into one number. The reason is empirical: in this study the components point in different directions—trace integrity passes cleanly while behavioural agreement fails, and composition improves on CartPole but collapses into conflict on Acrobot. A single aggregate would conceal exactly the heterogeneity that the failure-driven history was built to expose. We therefore define

$$\text{Auditability} = (\text{trace integrity, lossless coding, rule coverage,} \\ \text{behavioural agreement, composition quality, value-model reliability}),$$

and treat any claim that averages them as out of scope. This is the same six-item list as Table 1, and the predicates are fixed before the results are known; the verification mechanisms of Section 4.3 (hash-chain integrity, environment replay, provenance resolution) supply evidence for selected predicates without being predicates themselves. In particular, replay fidelity and provenance visibility are not interchangeable with composition quality and must not be substituted for it in the vector.

7.5 Limitations

The comparison is bounded in several ways. (1) The seed pool is $n = 8$; we report cross-episode uncertainty, not cross-seed uncertainty. (2) The single-actor baseline and the single-actor fallback ablation select an actor by training return mean, which is a post-hoc choice; on CartPole-v1 the selected seed (43) scores 163.65 on the matched evaluation set while four of the other seven seeds reach the 500 ceiling there, so the selection basis is demonstrably weak, and a different selection rule reverses the sign of the comparison (Table 8). (3) Environment ceilings make the CartPole comparison partially saturated (four of eight baselines already reach 500). (4) The confidence threshold and the arbitration rule were chosen after observing earlier failures; the two-setting threshold exploration is exploratory, not a systematic search. (5) The FQE-GPI baseline is approximate and off-policy; its failure is a diagnostic, not a general statement about GPI. (6) The behavior-agreement sample draws states from each actor’s own occupancy, so the pooled agreement is occupancy-weighted, not uniform over the state space. (7) Acrobot-v1’s random arbitration advantage was replicated over five random streams but the mechanism remains open. We disclose each of these rather than repairing them by assumption.

Representation and task-design limitations. The shared quantile symbolizer is itself a confounder: with only four bins per dimension, two policies can agree on a rule for the symbol while disagreeing on nearly every state inside it, so the rules describe the *symbol* and behavior lives on *states*. The communication intervention is a designed task and does not transfer to general RL agents; an early payoff design made communication unnecessary, a continuous encoder path could bypass the discrete codebook, and the blind receiver control had to truly block private information. The value baseline’s quality gate had to be pre-registered precisely because it can fail silently.

Future work. The unresolved mechanisms dictate the next steps: decompose the random-arbitration advantage by source rule, action frequency, and state occupancy; test return-weighted rule-bank induction under a pre-specified normalization and held-out evaluation; use the native sequence symbols as the control representation; and evaluate behavior on non-saturated tasks and across hardware.

Material gaps (honest disclosure). The replay and hash checks are operational-health assertions on the recorded runs. Cross-hardware replay is untested, the mechanism behind random arbitration remains undecomposed, and the communication result is bounded to the designed task.

7.6 Steelman

The strongest objection a reviewer can raise is this: *the headline result is small and the mechanism is untrustworthy. A +42.31 gain on a task where half the baselines already hit the ceiling—against a comparator chosen to be the weakest-looking one—produced by a fusion policy that is itself far below the ceiling and below the population mean, resting on one fallback-seed choice and an arbitration rule that is catastrophically worse than random on the second environment—this does not establish that rule-level fusion is a good idea.* We concede the weight of this objection and answer it with the evidence rather than by inflating the claim. The paper’s primary contribution is not the magnitude of the CartPole gain; it is (a) a protocol in which every rule is anchored to a replayed, hash-bound trace, so the description layer is operationally verified against the recorded artifact; (b) direct evidence that rule overlap does not imply behavioral agreement, which invalidates a common shortcut in the interpretability literature; and (c) an explicit, replicated negative result—confidence-ranked arbitration can be anti-selective under the evaluated conflict and occupancy regime—that no reviewer would have predicted from the CartPole results alone. The CartPole gain is real but bounded (paired, bootstrap interval excluding zero, matched episodes, post-hoc-selected comparator), and the Acrobot failure is reported in the same table that reports the gain, with the same measurement discipline. If the reviewer’s standard is “always-better composition,” we fail it, and the paper says so. If the standard is “auditable, bounded, falsifiable evidence about when rule composition helps,” the paper meets it.

Steelman, continued. The strongest reviewer objection is that the positive results are compatible with the rules contributing nothing beyond a state-conditioned action prior, with replay certifying only that the logger can reproduce its own numbers, and with the causal-intervention result living in a toy game whose decision-relevant payoff was engineered to make symbols useful. The response is evidence, not framing. First, the protocol record *is* the claim: the paper promises exactly replayable, provenance-visible composition under stated conditions and delivers the artifacts and the refutations that bound it. Second, the random-rule finding *constrains* rather than voids rule content: it is separated from action noise by a matched control (random-candidate -187.02 versus uniform-random action -498.71), which provides evidence that the candidate sets retain information beyond an unconditional uniform-random-action baseline—although that control does not isolate whether the information arises from rule content, state occupancy, action-frequency structure, candidate-set size, or the selection procedure—even though the tested deterministic resolver fails to exploit it. Third, the coordination task is disclosed as designed and decision-relevant by construction; its evidential role is to show that a causal-symbol test can pass *somewhere* with proper gates, not that symbols are generally causal. If the reviewer insists these bounds make the paper modest, the authors agree: modesty is the method.

7.7 Replicability note

All numbers above come from executed runs with hash-verified traces. The deterministic checks (replay, hashes, codec round-trip) are expected to reproduce exactly; the diagnostic statistics up to numerical precision; and the stochastic results (bootstrap intervals, random arbitration replicates)

within the reported intervals. Replaying a recorded output is not the same as generating a new one; the random-arbitration replicates are fresh generations under fixed streams.

Likely replication failures and their ratings. Three are foreseeable. (i) The arbitration result is sensitive to the selector stream: the initial stream gave -187.02 while five later streams ranged to -206.17 , so a single-stream replication would report a different magnitude. (ii) The Acrobot result depends on the return floor: with a median of -500.00 , a replication with different reset seeds could shift the mean without changing any qualitative conclusion. (iii) The behavioural-agreement result depends on the symbolizer, which is fitted to random-policy occupancy; a replication with a different calibration policy could move it.

Evidenced mitigations. Sharing one frozen symbolizer across seeds and arms within a task is HIGH—it removes the largest known confound in cross-seed comparison and is directly evidenced by the boundary differences of up to 0.157 observation units under per-seed fitting. Pairing all policy comparisons on identical reset seeds is HIGH—it is what makes the paired differences interpretable given the return variance. Reporting the decision mix alongside every fused return is MEDIUM—it prevents the most likely misreading. Repeating the random-selection condition across five independent streams is MEDIUM—it establishes the effect is not a single-stream artifact. For independent replication, the minimum requirement is the same frozen symbolizer, the same admitted rule banks with their support and confidence statistics, and the same evaluation reset-seed ranges.

Reproducibility Statement

All experiments use fixed seeds (control: 11, 29, 43, 71, 101, 149, 211, 307; composition comparisons: 100 matched reset seeds $20,000,000 + i$; communication: 6 independent runs). Rule induction uses 4 quantile bins per state dimension under a shared symbolizer, minimum support 8, minimum confidence 0.70. Composition comparisons are matched on reset seeds. Because deep RL results can vary substantially with random seeds, environment nondeterminism, evaluation protocol, and reporting choices [20], we report seed identities, matched reset seeds, evaluation distributions, and uncertainty intervals separately. The hash-chained trace logs, replay validator, and tamper-probe results are archived with the experiment records. Analyses not pre-registered are labeled exploratory. No human subjects, annotators, or sensitive data were involved.

8 Conclusion

We converted the question “what did the trained agents learn?” into an operationally verified rule-level description and composition protocol, and we measured where it works and where it breaks. Under a frozen shared symbolizer, eight independently trained agents can be described by comparable rule banks anchored to replay-verified, hash-bound traces; offline fusion of those banks exceeded the post-hoc-selected single-actor comparator on CartPole-v1 by $+42.31$ return points (95% interval [23.98, 61.93], exploratory because the comparator was chosen by training return and the task exhibits ceiling effects) while remaining auditable at the artifact level (replay and provenance checks pass under the evaluated setup). The same protocol fails on Acrobot-v1 in a way that isolates the fragile link—confidence-ranked arbitration is anti-selective under the evaluated conflict and occupancy regime, and uniformly random arbitration outperforms it. Rule overlap does not transfer to behavioral agreement, and a fitted-Q GPI baseline collapses in both environments. The failure-driven design history (lossy grammar compression, training-constraint interference, confidence-threshold collapse, serialization and ledger-schema defects) is reported as

evidence about which intuitions are unsafe. The honest summary is bounded: rule-level auditing is operationally verified under the evaluated replay, hashing, and provenance checks and is useful in low-conflict regimes; rule-level composition is environment-dependent; and the boundary conditions we identified are the paper’s main deliverable.

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A Per-Seed AIM Causal-Intervention Appendix

Table 12 reports the matched-token causal intervention per seed rather than only in aggregate, so that seed-level variation is visible; it is provided in the spirit of not relying on a single aggregate number. Each row is one independent run of the designed private-target communication game (Section 6.8). “Active position” is the message token whose parity mutual information is maximal; $\Delta P(B \mid C)$ is the *whole-message* change in the receiver’s action probability, whereas the matched-token contrast quoted in Section 6.8 is slightly smaller (six-run mean 0.99837, range 0.996986–0.999685); “shuffled” is receiver accuracy under a shuffled message; and “MI” is the active-token mutual information in bits against a permutation null. Values are recomputed from the archived run records of the six runs and are rounded for display.

The per-seed rows show that the aggregate is not carried by a single run: every seed attains $\Delta P(B \mid C) \geq 0.995$ with a shuffled-message control near chance (0.49–0.52) and an active-token MI of 0.98–1.00 bits. The active token position is not fixed across seeds (position 1 for four runs, position 0 for two), which is why the intervention is reported per position rather than as a single whole-sequence statistic.

Table 12: Per-seed matched-token causal intervention on the designed private-target game. Reported to expose seed-level variation. The scope is the designed task only and does not transfer to state-action rule fusion.

Seed	Active position	$\Delta P(B \mid C)$ (whole message)	Shuffled acc.	Active-token MI (bits)
0	1	0.9958	0.522	0.985
1	1	0.9977	0.503	0.992
2	0	0.9985	0.507	0.996
3	1	0.9961	0.496	0.994
4	0	0.9997	0.502	0.999
5	1	0.9994	0.491	1.000
Mean	—	0.9979	0.504	0.994